Sri Eshwar College of Engineering
Coimbatore | Tamilnadu
An Autonomous Institution
Affiliated to Anna University, Chennai

Department of Computer Science and Engineering

Course name : Data Science
Course code : U23AD491

Title of the project :
SMART HOSTEL ENERGY MANAGEMENT SYSTEM

Class / Section / Semester	:	III CSE / A / V
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Faculty I/C	:	Dr. M. Suriya, M.E., Ph.D., AP/CSE
Title	:	SMART HOSTEL ENERGY MANAGEMENT SYSTEM
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Abstract

AI-Powered Intelligent Court Case Prioritization and Judicial Assistance System

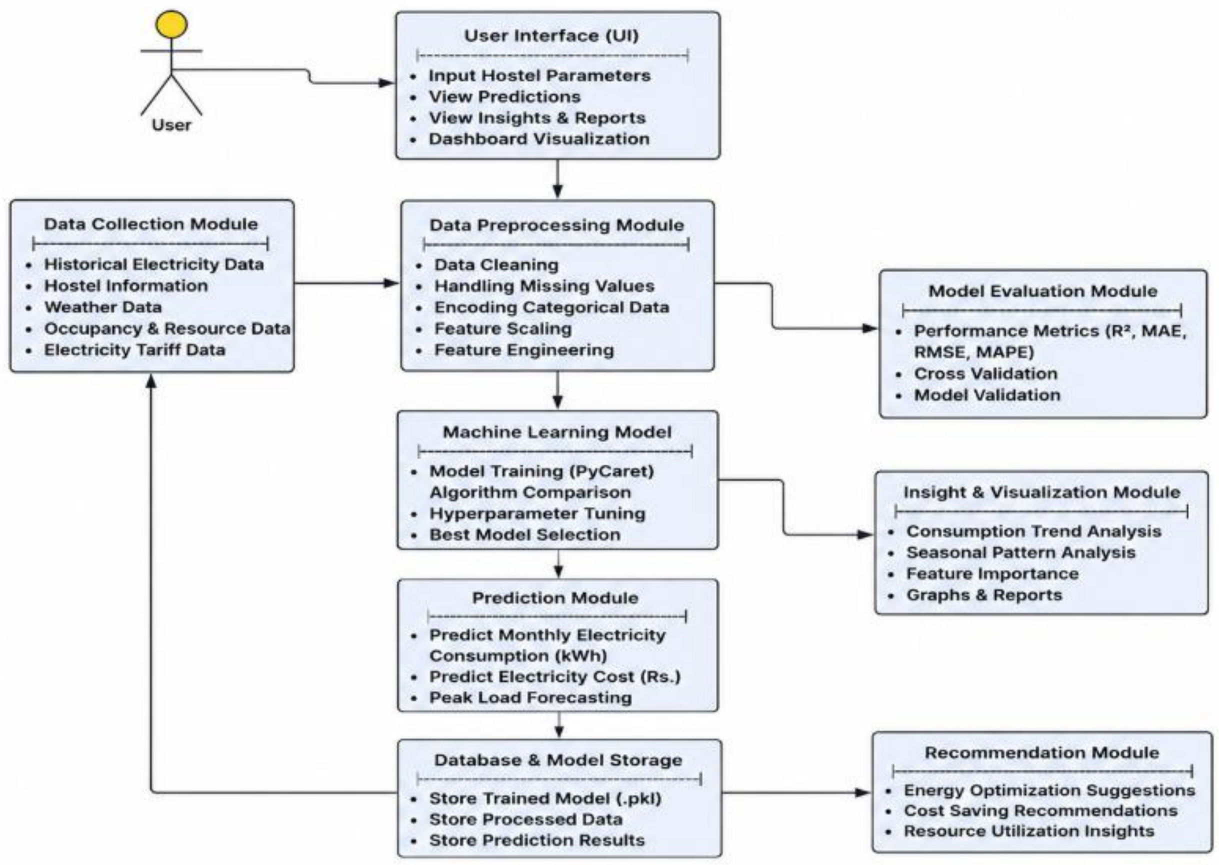
The judicial system faces significant challenges due to the growing backlog of pending cases, resulting in delays that can extend for several years or even decades. Judges, lawyers, and litigants often encounter difficulties in tracking case progress, managing procedural deadlines, identifying missing documents, and prioritizing cases based on urgency and legal significance. Existing court management systems primarily focus on storing case records and scheduling hearings but provide limited intelligent assistance for case prioritization and procedural guidance.

This project aims to address these challenges by developing an AI-powered Intelligent Court Case Prioritization and Judicial Assistance System that supports courts in managing cases more efficiently. The proposed system analyzes multiple factors such as case age, legal priority, urgency, case type, statutory deadlines, and procedural status to recommend an optimized order for hearing pending cases. Additionally, the system provides intelligent assistance by identifying required documents for each stage of the proceedings, notifying users of submission deadlines, suggesting when documents should be obtained from relevant authorities, and tracking procedural compliance throughout the case lifecycle. By integrating AI-driven prioritization, workflow monitoring, and decision-support capabilities, the proposed solution seeks to improve judicial efficiency, reduce case pendency, enhance transparency, and assist courts in delivering timely justice.

CONCEPT / SCOPE OF SOLUTION

- **Smart Hostel Energy Management System Using Machine Learning**
- Managing electricity consumption in hostels involves multiple challenges, such as monitoring energy usage, reducing operational costs, predicting future electricity demand, and promoting efficient resource utilization. Existing energy management systems often rely on manual monitoring or basic reporting tools, making it difficult for hostel administrators to identify consumption patterns, forecast future energy requirements, and implement proactive energy-saving measures. Additionally, fluctuations in occupancy, weather conditions, and equipment usage contribute to unpredictable electricity consumption.
- The Smart Hostel Energy Management System aims to address these challenges by providing an AI-powered solution that predicts monthly electricity consumption using machine learning algorithms. The system analyzes various factors such as hostel occupancy, number of electrical appliances, weather conditions, water consumption, solar energy generation, and peak power demand to estimate future electricity usage. Based on these predictions, the system assists hostel administrators in optimizing energy consumption, reducing electricity costs, improving resource planning, and supporting sustainable energy management through data-driven insights.

PROPOSED METHOD



Work flow diagram

TECHNOLOGY USED

- **Technologies Used**
- **Programming Language:** Python
- **Machine Learning:** PyCaret, Scikit-learn
- **Data Processing:** Pandas, NumPy
- **Data Visualization:** Matplotlib, Seaborn
- **Development Environment:** Visual Studio Code (VS Code)
- **Dataset Format:** CSV
- **Model Storage:** Pickle (.pkl)
- **Libraries:** Joblib, SciPy

PROJECT MODULES

- Project Modules
- Data Collection Module
- Data Preprocessing Module
- Feature Engineering Module
- Machine Learning Model Training Module
- Model Evaluation Module
- Prediction Module
- Data Visualization Module
- Energy Optimization & Recommendation Module
- Model Storage Module
- User Interface (Dashboard) Module

CONCLUSION

- **Conclusion**
- The **Smart Hostel Energy Management System Using Machine Learning** provides an intelligent approach to predicting monthly electricity consumption using historical and operational data. By applying machine learning algorithms, the system accurately analyzes factors such as hostel occupancy, weather conditions, electrical appliance usage, and resource consumption to forecast future energy demand. This enables hostel administrators to make informed decisions, optimize energy usage, reduce electricity costs, and improve overall resource management.
- The proposed system demonstrates how machine learning can support sustainable energy management through accurate predictions, performance analysis, and data-driven recommendations. In the future, the system can be enhanced by integrating real-time IoT sensor data, renewable energy monitoring, mobile applications, and cloud-based analytics to further improve prediction accuracy and support smart hostel infrastructure.